

Day 2: HTML Basics — Student Notes

Revision notes — key points, tables & examples

1. HTML Introduction

- HTML = HyperText Markup Language.
- Uses **tags** to structure content: `<tag>content</tag>`
- Not a programming language — only defines structure.
- Self-closing/void tags (no closing tag): `<br>` , `<img>` , `<hr>` , `<input>`
- Created by Tim Berners-Lee in 1991; current version is **HTML5**.

Block vs Inline Elements

Type	Behavior	Examples
Block	New line, full width	h1-h6, p, div, ul, li
Inline	Stays in text flow	a, span, img, b, i

2. Document Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  Content here
</body>
</html>
```

- **<head>** → invisible info (title, meta, links to CSS/JS)
- **<body>** → visible content

Useful head tags:

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="stylesheet" href="style.css">
```

3. Elements & Attributes

- Element = opening tag + content + closing tag
- Attribute = extra info in opening tag → `name="value"`

```
<a href="https://google.com" target="_blank">Google</a>
```

Common attributes: `href`, `src`, `alt`, `id`, `class`, `style`, `title`, `target`

id	class
Unique — only 1 per page	Reusable on many elements

4. Headings

```
<h1>...</h1> to <h6>...</h6>
```

h1 = biggest/most important → h6 = smallest. Use only one h1 per page.

5. Paragraphs & Text Formatting

```
<p>Text goes here</p>
```

Extra spaces in code are ignored. Use `<br>` for line break.

Tag	Effect
<code> / </code>	Bold
<code><i> / </code>	Italic
<code><u></code>	Underline
<code><mark></code>	Highlight
<code><small></code>	Smaller text
<code><sub> / <sup></code>	Subscript / Superscript

6. Lists

```
<ul><li>Item</li></ul>    → bullets
<ol><li>Item</li></ol>    → numbers
<dl><dt>Term</dt><dd>Definition</dd></dl>
```

`<li>` must be inside `<ul>` or `<ol>`. Lists can be nested inside each other.

ol attributes: `type="A/a/i/I"` , `start="5"`

7. Links

```
<a href="url">Link Text</a>
```

Type	Syntax
Email	<code>href="mailto:x@mail.com"</code>
Phone	<code>href="tel:+91..."</code>
Same page	<code>href="#id"</code>
New tab	<code>target="_blank"</code>

Relative path = same/nearby folder (about.html). Absolute path = full URL (https://...).

8. Images

```

```

`alt` = fallback/accessibility text (important!). Common formats: .jpg, .png, .svg, .gif, .webp.

Image as a link:

```
<a href="url"></a>
```

9. Comments

```
<!-- This is a comment -->
```

Not visible on the page. Used for notes, disabling code temporarily, or marking sections.

10. HTML Entities

Symbol	Entity
<	<

>	>
&	&
"	"
©	©
®	®
™	™
(space)	
₹	₹

11. Semantic Tags (Preview)

Tag	Use
<header>	Top section
<nav>	Navigation menu
<main>	Main content
<section>	Content grouping
<article>	Self-contained content
<footer>	Bottom section

12. Common Mistakes to Avoid

- Forgetting to close tags
- Wrong nesting order (close in reverse order)
- Missing quotes around attribute values
- Using more than one <h1>
- Typing raw < or > instead of entities
- Putting outside /

13. Void Elements — Quick List

, <hr>, , <input>, <meta>, <link>, <source> — none of these take a closing tag.

14. Writing Style Rules

- Tags/attributes aren't case-sensitive, but always write them in **lowercase** (industry standard).
- File names ARE case-sensitive on many servers — "Photo.jpg" ≠ "photo.jpg".
- Always close every non-void tag, even if the browser doesn't force you to.

15. Mini Glossary

Term	Meaning
Tag	Keyword in angle brackets, e.g. <p>
Element	Tag + content + closing tag
Attribute	Extra property inside opening tag
DOM	Tree structure browser builds from HTML
Void element	Tag with no closing tag/content
Nesting	Placing one element inside another

16. Extra Practice Snippets

Table of contents using links:

```
<ul>
  <li><a href="#intro">Intro</a></li>
</ul>
<h3 id="intro">Intro</h3>
```

Recipe using ordered list:

```
<ol>
  <li>Boil water</li>
  <li>Add tea leaves</li>
</ol>
```

Image gallery row:

```


```

Practical Reminder

Build a **Personal Profile Page** using: heading, image, paragraph, unordered list (hobbies), ordered list (skills), links (email/phone/social), a comment, and an HTML entity.